

SINGAPORE-MIT ALLIANCE

Computational Engineering CME5232: Cluster and Grid Computing Technologies for Scientific Computing COMPUTATIONAL LAB No. 1 5th July, 2006

INTRODUCTION TO HIGH PERFORMANCE COMPUTING RESOURCES

Objectives

To be familiar with the SMA Hydra Linux Cluster as the main High Performance Computing (HPC) platform for parallel computing tasks.

Background Knowledge

C/C++ programming languages, elementary knowledge of Unix/Linux commands.

- [Question 1](#)
- [Question 2](#)
- [Homework Problem to Prepare for Computational Lab 2](#)

Question 1: Getting to know the SMA Hydra Linux Cluster

SMA Hydra Linux Cluster (hydra.sma.nus.edu.sg) consists of 32 nodes:

- (a) 1 Access node: Pentium III 1GHz CPU x 2, 2GB memory
- (b) 31 Compute node: Pentium III 1GHz CPU x 1, 1GB memory

- How to access SMA Hydra Linux Cluster?

- (a) Windows: Install SSH client (Available at NUSNET-IV Web Shopping), SSH to **hydra.sma.nus.edu.sg** with your user ID and password
- (b) Linux: SSH is pre-installed, open a shell terminal and type **ssh hydra.sma.nus.edu.sg** with your user ID and password

Some Software/Libraries installed in the SMA Hydra Linux Cluster:

Application	Setting
Operating System	CentOS 4
GCC Compilers	GCC v3.4.3
Intel Compilers	C++ Compiler: /usr/local/intel_cc_81 Fortran Compiler: /usr/local/intel_fc_81 Intel debugger: /usr/local/intel_idb_81 eclipse: /usr/local/intel/eclipse
mpich v1.2.7	/usr/local/mpich/gnu (Default) /usr/local/mpich/intel mpicc: To use C MPI mpicxx: To use C++ MPI
Sun Grid Engine 6	
Condor v6.6.9	/home/condor
BLAST v2.2.11	/usr/local/BLAST
LAPACK	/usr/local/LAPACK Libraries: /usr/local/LAPACK/lib
SCALAPACK v1.1	/usr/local/SCALAPACK
BLAS	CBLAS: /usr/local/CBLAS/lib/LINUX BLAS (Fortran): /usr/local/BLAS/lib/LINUX
BLACS v2.2.11	/usr/local/BLAST
PETSc v2.3.1 (Full version - updated)	/usr/local/PETSc/petsc-2.3.1
Java jdk-1.5.0-03	/usr/local/jdk1.5.0_03

Go to <http://hydra.sma.nus.edu.sg> for more information about SMA Hydra Linux Cluster

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Question 2

It is required to evaluate the integration of a continuous function approximated by using the trapezoidal rule. Sample serial program **ori_integral.c** is provided.

- How to locate and copy the program **ori_integral.c**?

The CME5232 course directory in SMA Hydra Linux Cluster is **/home/cme5232**.

All the sample program code will be stored in **/home/cme5232/clab/clabx** (where **x** is either 1, 2, 3, 4, 5 or 6)

Your account are given the **read** and **execute** permission for the CME5232 course directory so that you can read and copy the file.

The following is a sample procedure to copy the given sample program from the CME5232 course directory:

- (1) **mkdir cme5232** in your home directory and **cd cme5232**
- (2) **mkdir clab** and **cd clab**
- (3) **mkdir clab1** and **cd clab1**
- (4) **mkdir qn2** and **cp /home/cme5232/clab/clab1/qn2/* qn2**
- (5) **cd qn2**

- How to compile the C/C++ program?

C : **gcc ori_integral.c -o ori_integral** (Note: **-lm** option is required if function in **<math.h>** is used)

C++: **g++ ori_integral.c -o ori_integral**

- How to compile the C/C++ program with optimization level (O1, O2, O3)?

gcc ori_integral.c -o ori_integral -Ox (where **x** is either 1, 2 or 3)

- How to run/execute the compiled C/C++ program?

./ori_integral

- How to abort the program execution in case it is required? (Not recommended unless neccessary)

Press **Ctrl-C**

- How to check if the process is successfully removed?

ps -ef | grep user ID (e.g. my **user ID** is g0501972)

- What if the job is not successfully removed?

kill -9 pid (where **pid** is the process id)

- It is NOT recommended to execute the program on the submit node of a cluster, what should we do?

Submit the job using **Sun Grid Engine** (Latest Version: [N1 Grid Engine 6](#)).

SGE is Sun's resource management solution. It provides the power and flexibility required for campus grids...

[Read more](#)

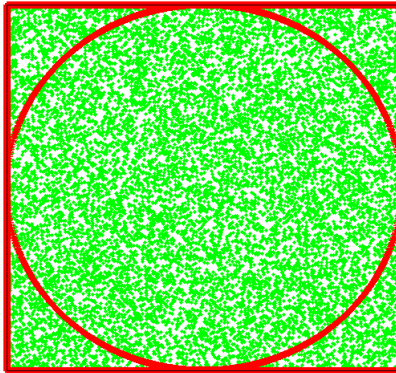
A resource management software is extremely important in computation-intensive environment, especially when MPI program is required.

In SMA Hydra Linux Cluster, job submission via SGE is compulsory. The job submission will be introduced in the Computational Lab 2.

Homework Problem to Prepare for Computational Lab 2

Students are required to choose one out of two following methods to compute the value of π .

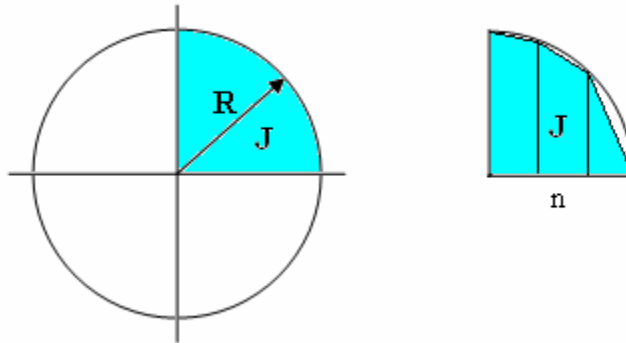
Monte Carlo method



It is required to use the Monte Carlo simulation method to compute the value of π . This method consists of inscribing a circle inside a square and then throwing random "darts" at the square. The ratio of "darts" that land inside the circle (n) to the total number of the darts thrown (N) will be ratio of the area of the circle to the area of the square ($\pi/4$). From this, the value of π can be calculated from $\pi = 4 \cdot n/N$.

You are expected to write to write out the sequential version of the code. Run the sequential code by varying the total number of the "darts" thrown. Check the precision of the value of π .

Continuous method



It is required to use trapezoidal rule to evaluate a quarter area of the circle with radius R . Then the value of π can be calculated by 4 times ratio of the quarter area of the circle to the square of the circle's radius (R^2).